

SAI SRILEKHA ALURU

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Experience

Clinical Informatics Data Analyst

Indiana Institute on Disability and Community - Hands In Autism

January 2025 – December 2025

Bloomington, IN

- Reviewed and validated outpatient EHR data across scheduling, encounters, and clinical documentation to ensure data accuracy, completeness, and regulatory alignment.
- Cleaned and mapped clinical data elements across workflows to support consistent reporting, improving overall reporting reliability by 15%.
- Designed and maintained Power BI dashboards that monitored documentation completeness, system usage patterns, and workflow trends, used by operational teams during weekly reviews.
- Produced recurring clinical and operational reports using Excel and Python, translating raw system data into actionable summaries for program leadership.
- Maintained detailed documentation of data workflows, validation logic, and issue resolution steps to support SOP compliance and continuity across system updates.

Research Assistant

Indiana University – Luddy School of Informatics, Computing, and Engineering

September 2024 – December 2025

Indianapolis, IN

- Prepared clinical research datasets by performing structured data cleaning, validation, and consistency checks prior to analysis.
- Used Python to preprocess and transform structured clinical datasets, including handling missing values, standardizing variables, and preparing analysis-ready files.
- Aligned variables across multiple data sources using Excel and SQL to support accurate and reproducible statistical analysis.
- Generated summary tables and analytical outputs for faculty investigators to support study interpretation and reporting.
- Documented data handling methods and transformation steps, contributing to an estimated 20% reduction in recurring data preparation time for follow-on analyses.

Research Data Analyst

Amura Health Private Limited

November 2022 – October 2023

Chennai India

- Worked with clinical trial datasets across phases I–IV, ensuring alignment with GCP requirements throughout the data lifecycle.
- Reviewed subject-level data against Case Report Forms (CRFs) to identify discrepancies, missing entries, and protocol deviations.
- Structured clinical datasets using CDISC (SDTM) standards to support standardized analysis and downstream reporting.
- Conducted statistical analysis in R, including descriptive statistics and trend analysis, to identify data inconsistencies and clinical patterns.
- Executed data preparation and reporting activities in line with SOPs and regulatory requirements, reducing internal data clarification queries by 30% during review cycles.

Clinical Pharmacist

Aster CMI Multi-Speciality Hospital

October 2021 – October 2022

Bengaluru, India

- Acted as a Cerner Millennium super user, supporting inpatient medication management and clinical documentation workflows.
- Reviewed medication orders, patient charts, and electronic records to ensure data accuracy and continuity of care.
- Monitored system alerts and application behavior, escalating recurring issues to informatics and IT teams for resolution.
- Developed medication utilization and operational reports using Cerner Discern Analytics to support quality improvement initiatives.
- Assisted with order set testing, validation, and workflow review following system updates, contributing to a 25% reduction in medication-related documentation issues.

Education

Indiana University

Master of Science in Health Informatics | GPA:3.8/4.0

KLE College of Pharmacy

Doctor of Pharmacy | GPA:.8.2/10

Indianapolis, IN

January 2024 – December 2025

Bengaluru, India

August 2016 – September 2021

Skills

- **Programming:** Python, R Shiny, SAS, SQL (MS SQL)
- **Data Visualization and Reporting:** Power BI, Tableau, GraphPad Prism
- **Healthcare Data Standards & Interoperability :** CDISC, FHIR, ICD-10, SNOMED, LOINC, RxNorm, HL7
- **Machine Learning & NLP:** Predictive modeling, NLP pipelines, sentiment analysis
- **Healthcare & Regulatory:** Good Clinical Practice (GCP), ICH guidelines, OnCore, HIPAA compliance, Pharmacovigilance
- **Clinical Data Management:** Data validation, Quality control (QC) logic, Data capture form design, Data Abstraction, Abstraction guidance, Medical record abstraction, Case Report Forms (CRFs), Outcomes reporting, Data audits
- **Statistical & Analytical Methods :** Descriptive statistics, survival analysis, predictive modelling
- **Clinical Systems & Databases:** Electronic Health Records (EHR), Electronic Data Capture (EDC) systems, EMRs, Clinical databases, OnCore, Cerner

Projects

- **ETL Pipeline for FHIR API Integration:** Designed and implemented a Python-based ETL pipeline to extract, transform, and load healthcare data from FHIR APIs, mapping clinical information to core FHIR resources to ensure semantic consistency. Used FHIR RESTful APIs to enable standardized, interoperable access to clinical data across heterogeneous systems, validating data integrity and terminology alignment with LOINC, SNOMED CT, and RxNorm. Automated transformation workflows resolved cross-system inconsistencies and improved data reliability for analytics-ready datasets. Visualized longitudinal patient trends using R and built an R Shiny web application to interactively explore patient-level and aggregate clinical metrics, supporting informatics-driven insights
- **COVID-19 Vaccination Impact Analysis:** Evaluated large-scale COVID-19 datasets using R to assess vaccination coverage and trends in patient outcomes across populations. Machine learning–based exploratory modeling techniques were applied to identify patterns and associations between vaccination status and outcome measures, supporting population-level insight generation. Results were communicated through clear analytical visualizations in R, enabling interpretation of coverage patterns and outcome trends for reporting purposes.

Publication

“Analysis of Prescription Pattern of Antimicrobial Agents in General Medicine in a Tertiary Care Hospital” – Conducted 6-month study of 150 inpatients at Aster CMI Hospital; analyzed 185 prescriptions; found cephalosporins (54.8%) most prescribed for UTIs and RTIs; documented one ADR; results supported antibiotic stewardship.

Link to the publication : https://wjpr.s3.ap-south-1.amazonaws.com/article_issue/d6c12b3b38a2432e851bc895b11cf566.pdf